Nistala Sai Akhil

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.50
Fiitjee Junior College	Class XII (TSBIE)	2023	96.40%
Bharatiya Vidya Bhavan's Public School	Class X (CBSE)	2021	96.60%

WORK EXPERIENCE

Software Development (SDE) Intern | Mitsubishi Electric, Japan

- Developed a full-stack warehouse management platform using **Next.js, React and MySQL** to monitor drone feeds, manage inventory, visualize rack layouts, and track warehouse operations through a unified real-time dashboard.
- **Built REST APIs and WebSocket-based real-time communication** for live video streaming and data ingestion, enabling automatic UI synchronization and continuous warehouse inventory updates without manual page refreshes.
- Designed **modular backend services and database workflows** for drone-driven warehouse operations, implementing QR-based inventory ingestion and automatic rack assignment with clear separation across UI, API, and data-access layers.

PROJECTS

CodeHarbor | Version Control & Repository Management Platform [Github]

- Built a full-stack repository platform exposing **20 REST endpoints** for auth, repositories and issues with JWT security.
- Designed **3 MongoDB/Mongoose schemas** with compound indexing to enforce per-owner repository uniqueness.
- Developed a **6-command Git-like CLI** with AWS S3-backed commit storage and remote synchronization via AWS SDK v3.
- Implemented ownership-based authorization and **repository visibility controls across 20 API endpoints**.

MeetSphere | Video Conferencing Platform [Github]

- Built a full-stack conferencing platform supporting **video calls, screen sharing, live chat and persistent meeting history**.
- Engineered **peer-to-peer WebRTC communication** with Socket.IO signaling to enable real-time multi-participant calls.
- Implemented **7 real-time events** for signaling, presence, chat and media-state, enabling consistent room state.
- Designed **4 MongoDB schemas** with indexed session data, enabling persistent participant tracking and paginated history.

AdLens | Agentic Performance Investigator [Github]

- Built a **6-stage stateful LangGraph workflow** orchestrating **11 SQL/analytics tools** across **45.6K MySQL records**.
- Engineered **robust schema validation, parameterized SQL, and structured tool execution** across **153 end-to-end trials**.
- Built a **deterministic statistical validation engine** achieving **94.1% anomaly recall** with **zero false alarms** on clean days.
- Added **local Ollama inference** and a **240K-row MySQL workload** spanning **20K restaurants**.

RESEARCH PUBLICATION

N.S. Akhil, P. S. Nishant, U. V. Rohith, and **S. Upadhyaya**, "**Spatio-Temporal Reconstruction of MODIS Land Surface Temperature over Hyderabad Using ConvLSTM**", IEEE India Geoscience and Remote Sensing Symposium (InGARSS 2026)

SKILLS

- **Programming:** C, C++, Python, JavaScript, TypeScript, Object-Oriented Programming (OOP), Data Structures & Algorithms
- **Frontend:** HTML5, CSS3, React.js, Next.js, React Router, Redux, Tailwind CSS, Bootstrap, Material UI, State Management
- **Backend:** Node.js, Express.js, REST APIs, MVC, Middleware, Authentication & Authorization, API Integration
- **Databases:** MongoDB, MySQL, Database Design, CRUD Operations, SQL Queries, Database Management (DBMS)
- **AI/GenAI:** OpenAI, Claude, LLMs, RAG, Prompt Engineering, LangChain, LangGraph, Vector Databases, Semantic Retrieval
- **Tools & Technologies:** Git, GitHub, Postman, Jest, WebSockets, WebRTC, Socket.IO, API Testing, Jira, Linux, Scrum, Confluence
- **DevOps & Cloud:** Docker, Kubernetes, CI/CD, AWS, Cloud Deployment, Containerization, Cloud-Native Development

POSITIONS OF RESPONSIBILITY

Overall Head | International Relations Cell, IIT Hyderabad

- **Spearheaded** Japan Day with **10+ Japanese universities**, managing itineraries, communication, and multi-team execution.
- **Directed** visits for **50+ international students across 12+ countries**, managing accommodation, transport and schedules.
- **Established 25+ MoUs across 15+ countries**, driving university outreach, documentation, follow-ups, and approvals.

Student Head Boy | Bharatiya Vidya Bhavan's Public School

- **Commanded** a 15-member student council, delegating responsibilities across school-wide activities and initiatives.
- **Advocated** for the student body as the direct link to the **Principal**, raising concerns and driving follow-ups.
- **Executed** student-led events, delegating responsibilities and managing teams through completion.

Class Representative (CR) | Civil Engineering '23 Batch

- **Represented** 60+ students in faculty discussions, consolidating batch-level academic issues into structured requests.
- **Coordinated** course updates, deadlines, timetable changes, and faculty communications for a **60+ student cohort**.
- **Resolved** student-faculty issues by consolidating feedback and driving **timely follow-ups to closure**.

EXTRACURRICULAR

- **Theatre Arts:** Member of **Swaang, Rang De Manch, IIT Hyderabad**, contributing to stage productions and performances.
- **Hindustani Classical Keyboard:** Practicing for **5+ years**, performing in cultural events and college activities.
- **State-Level Cricket:** Represented Medak district at **U-17 level**, competing in state-level cricket tournaments.
- **Hackathon Finalist, Pulse Quest**, a competitive innovation challenge conducted by **KLA, Deutsche Bank and Elan nVision**